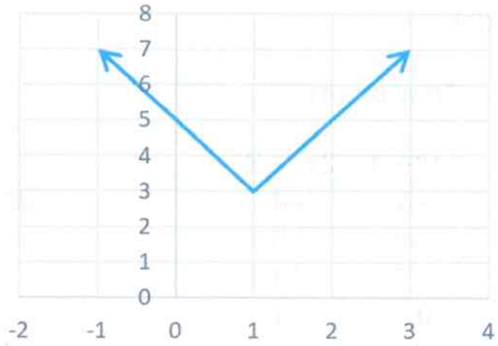
Example: Sketch $f(x)$ Using Key Points

Sketch the graph $f(x) = 2|x-1| + 3$ using key points.

The parent function is $f(x) = |x|$. Recall the key points of this function.

Original points: New points:

x	y	$x+1$	$2y+3$
-2	2	-1	7
-1	1	0	5
0	0	1	3
1	1	2	5
2	2	3	7



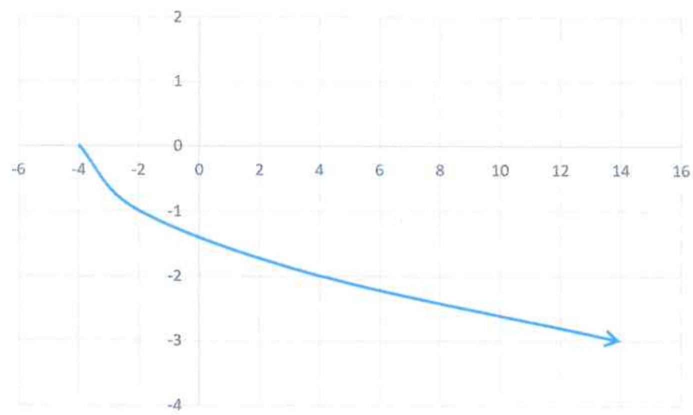
Example: Sketch $f(x)$ Using Key Points

Sketch the graph of $f(x) = -\sqrt{\frac{1}{2}(x+4)}$ using key points.

The parent function is $f(x) = \sqrt{x}$. Recall the key points for this function.

Original points: New points:

x	y	$\frac{x}{\frac{1}{2}} - 4$	$-y$
0	0	-4	0
1	1	-2	-1
4	2	4	-2
9	3	14	-3



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Worksheet – Parent Functions

1. (KU – 16 marks) Describe the transformations for each function and sketch a graph of each using five key points. Also, state the domain and range.

a. $f(x) = |x - 2| + 1$ b. $f(x) = -\frac{2}{x+1}$ c. $f(x) = \sqrt{2x + 6} - 2$

d. $f(x) = -2(x - 1)^2 + 3$

2. If $f(x)$ and $g(x)$ are equivalent, what must a be?

$f(x) = \frac{1}{\frac{1}{3}x - 2} + 5$ $g(x) = \frac{a}{x-6} + 5$

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Quiz 2 – Graphing Parent Functions

1. (KU – 20 marks) For each function, describe the transformations done, use 5 key points to graph the function. Also, state the domain and range, and intervals of increase and decrease.

a. $f(x) = \frac{2}{x+1} - 3$ b. $f(x) = -\sqrt{2x+2} + 3$ c. $f(x) = 3|x+2|$

d. $f(x) = 2(x+1)^2 - 2$

2. (TI – 4 marks) Find the inverse of the following function, and graph $f^{-1}(x)$, restrict the domain of $f(x)$ so that $f^{-1}(x)$ is a function.

$$f(x) = x^2 + 4x - 1$$

3. (TI – 1 mark) If $f(x)$ and $g(x)$ are equivalent functions, what must a be?

$$f(x) = \sqrt{9x-18} - 1 \qquad g(x) = a\sqrt{x-2} - 1$$

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